PAL3 Task Design Document

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1.0	Kareem Zaghloul, John Wittig, Robert Yaffe, Timothy Sheehan, Stas Busygin	First version; Adapted from FR3 by Youssef Ezzyat	05/03/2016

Overview

The PAL3 task implements closed loop brain stimulation during a memory task to test the hypothesis that direct electrical stimulation (DES) of the brain during poor memory states results in better memory than a control condition with no brain stimulation. This work builds on two prior experiments: PAL1 and PAL2.

In PAL1 we recorded brain activity from multiple implanted electrodes during the encoding and retrieval of word pair lists and identified neural correlates of good and poor memory encoding states (henceforth biomarkers). We sought to predict whether a given item would be recalled based on the spectral decomposition of the time-series of bipolar neural recordings from all N channels recorded. We thus fit an L2-regularized logistic regression model to the item level data in which the LHS variable was the recalled/non-recalled status of an item and the RHS variables were the wavelet power in each of 8 frequency bands and across all N channels (henceforth classifier). The model was fit to data from sessions 1 through M-1 and validated on session M, with the penalty parameter determined based on the distribution of best fitting parameters across all patients who performed the experiment excluding the patient whose data were being fit. The model was able to reliably predict recall based on the spectral decomposition of the encoding period data.

The present study (PAL3) aims to directly test the hypothesis that DES delivered during poor memory states can produce reliably better memory than observed on control lists without any DES. On half of the experiment lists, DES will be under control of an algorithm that evaluates the state of the brain for each presented word pair and decides whether or not to stimulate vs. control lists in which no stimulation is provided.

Methods

Except as noted, the design of the behavioral task and the procedure for it are identical to PAL1. The reader is referred to that design document for further details. Subjects will perform paired associates learning on 26 study-test lists. Stimulation is not delivered to the brain during lists 1-4 as these lists are used to evaluate the ability of the logistic-regression classifier, trained on the same patient's PAL1 data, to predict the recalled status of items in PAL3. Following the first 4 lists, the remaining 22 lists will be equally divided into two conditions: stimulation during encoding (StimEnc) and no stimulation (NoStim) lists. Studies investigating stimulation during retrieval (StimRet) will be conducted at a later time. These 22 lists will be pseudorandomly interleaved to balance the number of StimEnc and NoStim lists in the first and second halves of the session.

ECoG/iEEG Analysis

During stimulation lists, EEG data are analyzed to determine when to turn on the stimulator. The raw EEG data is converted from the default referential recording to a bipolar reference between neighboring electrodes as determined by the bipolar montage stored in each patient's

'*Talairach structure*'[1]. These bipolar recordings are then filtered to remove 60Hz line noise using a Butterworth stop band filter (58-62Hz range, fourth order). A Morlet wavelet transformation is then applied to the signal (wave number = 5) to estimate spectral power during word pair encoding periods (8 wavelets log-spaced between 3-180Hz; center frequencies: [3.0, 5.4, 9.7, 17.4, 31.1, 55.9, 100.3, 180] Hz). For analysis of both the training (PAL1) and closed-loop data (PAL3) we use power from 300-2000 ms after word pair onset with a mirrored buffer pre- and post- (1000 ms buffers; total required data window = 3700 ms). Given our lowest frequency wavelet (3 Hz), the minimum required buffer size to avoid convolution edge artifacts is given by one half of the length of the wavelet = 929 ms.[2]

After applying the wavelet transformation we log-transform raw power values. These log transformed power values are then z-transformed to normalize them based on the within-session mean and standard deviation across all word pair encoding events in the session. The resulting z-transformed log power values are then features for the pattern classifier described below.

L2-logistic regression classifier: Prior to the start of PAL3 we will have fit a regression model to data from multiple PAL1 sessions as described below. This regression model will determine the timing of stimulation in PAL3 stim lists. Using the spectral features described above we will estimate an L2-penalized logistic regression model to predict whether an item will be subsequently recalled (a proxy for good memory). Before estimating the beta weights (coefficients) on the input features one must specify a penalty parameter to be used in the estimation routine. We determine the penalty parameter based on the penalty value that leads to the highest average AUC in all prior RAM subjects with at least two PAL1 sessions. The penalty parameter will be inversely weighted according to the class imbalance (Hastie et al., 2001). Class weights for each class are computed as $(1/n_A)/((1/n_A + 1/n_B)/2)$ where n_A is the number of events class A events, and n_B is the number of class B events. This ensures best fit values of the cost parameter, C, are comparable across different class distributions. For example, if the proportion of observations in classes A/B are .25 and .75 respectively, then the ratio of the weights for the penalty parameters for each class will be 3:1. The weight for class A will be .25/((.25+.75)/2) = .5, and the weight for class B will be .75/(.25+.75)/2 = 1.5. The penalty parameter will be the value that leads to the highest average AUC in all prior RAM subjects who had at least two PAL1 sessions with at least 40 lists in total.

Criterion for progressing from PAL1 to PAL3. Each patient's classifier will be tested for out-of-sample performance using leave-one-session-out (LOSO) cross validation. The true LOSO AUC will be compared to an AUC distribution (200 samples) generated from classification of label-

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[1] *Talairach structure* refers to the eeg_toolbox/PTSA data object used by the RAM team to represent information about each patient's montage. For the purposes of PAL3, the relevant information is the set of bipolar electrode pairs, not the coordinate system used to represent the electrode spatial locations. The structure in fact contains electrode information in Montreal Neurlogical Institute (MNI) space; the 'Talairach' nomenclature has nonetheless been maintained from earlier versions of the toolboxes.

[2] One half of the length of the wavelet is given by $n_z s f_s$ where n_z = number of standard units on the Gaussian; s = standard deviation of the Gaussian wavelet window; and f_s = sampling frequency.

permuted input data. P < 0.05 one-tailed will be used as the significance threshold for continuing to PAL3. At least three sessions with a minimum of 15 lists per session are required. If classifier is not significant with three sessions, additional PAL1 sessions should be collected.

Closed-loop Procedure

During the item presentation phase of the StimLists in PAL3 we will evaluate the classifier output during the 300-2000ms interval following word pair onset to decide whether to apply stimulation intended to affect the current word pair (this epoch will be mirror buffered by 1000ms on each end). If the classifier output during this interval is below the median of the classifier output in PAL1 sessions, we will stimulate immediately. This metric is stable and should maximize the accuracy of our decision.

Data Normalization Process. The values that are input to the real-time classifier will be normalized to ensure the range of the data match the range expected by the classifier model. This will be done using the mean and standard deviation across word pair encoding events in the first four lists (one practice list and three NoStim lists). Mean and standard deviation estimates will be updated as more NoStim lists are collected. Updates will occur after every NoStim list (following the recall period). To do so, StimControl.m will need state information about timing of word pair onset.

Classifier Validation. To validate the classifier for use in PAL3, the output from the classifier for each item presented on the first four lists will be compared to the classifier output when tested on data from all previous sessions using a two-sample Kolmogorov-Smirnov test. The null hypothesis that the PAL3 and training data come from the same distribution must not be rejected (i.e., P > 0.05) for the PAL3 task to continue. If this hypothesis is rejected, an error will be thrown and the session will proceed as a PAL1 session. StimControl.m will also output (to the command line and to a log file on disk) the results of the two-sample Kolmogorov-Smirnov test comparing the distributions.

Stimulation Artifacts. We will avoid decoding brain data during stimulation. We will accomplish this by stimulating only after our decoding period and only during the given word pair. The ISI will leave plenty of time for stimulation artifacts to subside before the next decoding period.

Stimulation Onset Timing. A stimulation waveform, specified by the Parameter Search (PS) task (see below), will be triggered when the multivariate classifier detects that the classifier output is below threshold. Specifically, stimulation decisions will be made based on classifier decoding of the word pair presentation interval. If the classifier output for the decoded word pair falls below threshold, a stimulation waveform will be triggered as described in the methods summary.

Selection of Closed Loop Stimulation Parameters

We will find the parameter set, $\vec{\theta}$ across all PS sessions that maximizes the t-statistic of the post-pre expected recall performance change against zero. Based on the parameter manipulated across PS2, PS2.1, and PS3 tasks we expect to identify a set theta that includes:

- Stimulation location (bipolar electrode pair)
- Amplitude **(SafeAmp-.5, SafeAmp-.25 or SafeAmp mA)**
- Pulse frequency **(10, 25, 50, 100, 200 Hz or single pulse)**
- Theta pattern frequency **(3, 4, 5, 6, 7, 8 Hz or n/a)**
- Theta train frequency **(100, 200 Hz or n/a)**

In the absence of patient-specific PS data, $\vec{\theta}$ will be chosen according to the best group-average parameters from the vault of PS datasets in existence at the time of the patient's participation.

Other Stimulation Parameters:

- Based on the aggregate PS1 data pulse duration will be set to 500ms
- Bipolar stimulation
- Pulse width: 655 microsecond pulse (300 us per phase, 55 us interphase)

References

- Hastie. T., Tibishirani, R., & Friedman, J. (2001). The elements of statistical learning: data mining, inference and prediction. *New York: Springer*.